

DIMITRI CHAKMA

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Summary

Python developer who designed and shipped two production-deployed agentic AI systems end to end, from architecture through Docker and Railway deployment. Built a CBT and mental-health RAG agent with hybrid Neo4j graph and Pinecone vector retrieval, a self-correcting retrieval router, and a crisis-safety gate, alongside a habit-coaching agent using LangGraph, retrieval-augmented memory, and an LLM-as-a-Judge evaluation suite. Backend foundation in FastAPI, PostgreSQL and pgvector, and JWT authentication, backed by Machine Learning, Deep Learning, and NLP specializations from DeepLearning.AI.

Education

United International University M.Sc. in Computer Science & Engineering, Major in Data Science (Ongoing)	2024 – Present
Ahsanullah University of Science & Technology B.Sc. in Electrical & Electronic Engineering, CGPA: 3.084	2023
Government Rajendra College Higher Secondary Certificate (HSC), GPA: 4.42	2017

Technical Skills

- Programming Languages:** Python, JavaScript, TypeScript, R
- Generative AI & Agentic Systems:** LangChain, LangGraph, RAG (Pinecone, pgvector), Knowledge Graphs (Neo4j), LLM-as-a-Judge Evaluation (Ragas, DeepEval), MCP
- Backend & APIs:** FastAPI, REST API design, JWT authentication, SQLAlchemy
- Databases & Tools:** PostgreSQL, pgvector, Neo4j, Pinecone, SQL, SQLite, Git
- Cloud & DevOps:** Docker, Railway, Streamlit Cloud, LangSmith (tracing), pytest
- BI & Data Tools:** Microsoft Power BI, Excel
- Languages:** Bangla, English, Hindi

Projects

- AI Habit Tracker** | Personal project | [GitHub](#)
- Engineered a habit-coaching agent (FastAPI, LangGraph, Claude) with retrieval-augmented long-term memory, embedding weekly behavioral summaries in pgvector to ground every recommendation in real user history.
 - Built an LLM-as-a-Judge evaluation suite (DeepEval, pytest) across faithfulness, context precision, and a custom empathy metric, catching two real regressions, including an internal tool-name leak, before release.
 - Layered safety guardrails, including prompt-injection and harmful-content classification, PII masking, per-user rate limiting, and token budgets, on top of JWT authentication with per-user data isolation.
 - Deployed and operated the full system in production: Dockerized backend on Railway, Streamlit Cloud frontend, Neon PostgreSQL with pgvector, a Telegram bot via webhook, and end-to-end LangSmith tracing.
 - Stack: Python, FastAPI, LangChain/LangGraph, PostgreSQL/pgvector, Docker, Railway, DeepEval, LangSmith, JWT, MCP.
- CBT & Mental Health Companion (RAG Agent)** | Personal project | [GitHub](#) | [Live Demo](#)
- Built a production-shaped RAG agent (LangGraph) that answers CBT and mental-health questions strictly from a curated knowledge base, using hybrid Neo4j graph and Pinecone vector retrieval.
 - Designed a corrective-retrieval router that classifies each sub-question, grades its own retrieval quality, and falls back through graph, vector, and live web search before ever answering "I don't know."
 - Built a safety-first request path with a front-door crisis-risk gate that classifies bare follow-ups in context, routing to IP-resolved country-specific crisis helplines with an international fallback.
 - Implemented defense-in-depth auth across two independently deployed services, bridging Google OAuth identity from the Next.js frontend to the FastAPI backend via a short-lived signed token, with server-side ownership checks on every lookup.
 - Engineered cost and abuse controls, including a shared rate limiter, Postgres-backed response caching, a hard daily spend ceiling, per-IP limiting, and prompt-injection hardening on all user- and web-sourced text.
 - Built a two-layer evaluation system (custom harness plus Ragas) that scored 93% overall pass rate on a 45-item golden set, with 100% on safety-gate correctness, graph accuracy, and multi-turn memory.

- Stack: Python, TypeScript, LangGraph, FastAPI, Next.js, Neo4j, Pinecone, Claude (Anthropic), Postgres, Auth.js/OAuth, Ragas, Docker, Railway.

Certifications & Training

- Machine Learning Specialization — DeepLearning.AI, Coursera
- Deep Learning Specialization — DeepLearning.AI, Coursera
- Natural Language Processing Specialization — DeepLearning.AI, Coursera
- Data Analysis with Pandas & Python — Udemy
- Microsoft Power BI — Udemy
- Data Analytics — Bohubrihi